Summary and Highlights

ACID stands for Atomicity, Consistency, Isolated, and Durable.

- BASE stands for Basic Availability, Soft-state, and Eventual Consistency.
- ACID and BASE are the consistency models used in relational and NoSQL databases.
- Distributed databases are physically distributed across data sites by fragmenting and replicating the data.
- Fragmentation enables an organization to store a large piece of data across all the servers of a distributed system by breaking the data into smaller pieces.
- You can use the CAP Theorem to classify NoSQL databases.
- Partition Tolerance is a basic feature of NoSQL databases.
- NoSQL systems are not a de facto replacement for RDBMS.
- RDBMS and NoSQL cater to different use cases, which means that your solution could use both RDBMS and NoSQL.

Availability	In the context of CAP, availability means that the distributed system remains operational and responsive, even in the presence of failures or network partitions. Availablity is a fundamental aspect of distributed systems..
BASE	An alternative to ACID. Stands for Basically Available, Soft state, Eventually consistent. BASE allows for greater system availability and scalability, sacrificing strict consistency in favor of performance.
Basically available	A basically available system remains operational even in the presence of failures or faults.
CAP	CAP is a theorem that highlights the trade-offs in distributed systems, including NoSQL databases. CAP theorem states that in the event of a network partition (P), a distributed system can choose to prioritize either consistency (C) or availability (A). Achieving both consistency and availability simultaneously during network partitions is challenging.
Consistency	In the context of CAP, consistency refers to the guarantee that all nodes in a distributed system have the same data at the same time.

Consistent	The action of being consistent ensures that a database transaction transforms the database from one consistent state to another.
Durable	Guarantees that once a transaction is committed, its changes are permanent and will survive any system failures.
Eventually consistent	An eventually consistent system reaches a consistent state, where all nodes have the same data given that there are no new updates.
Isolated	Isolation refers to the property that multiple transactions can run concurrently without affecting each other.
Partition tolerance	In the context of CAP, partition tolerance is the ability of a distributed system to continue functioning even when network partitions or communication failures occur.
Replication	Replication involves creating and maintaining copies of data on multiple nodes to ensure data availability, reduce data loss, fault tolerance (improve system resilience), and provide read scalability.
Sharding	Refers to the practice of partitioning a database into smaller, more manageable pieces called shards to distribute data across multiple servers. Sharding helps with horizontal scaling.